

Heat Transfer Exam 1

$$q''_{cond} = -k \frac{dT}{dx} \quad q''_{conv} = \bar{h}(T_s - T_\infty) \quad q''_{rad} = \Sigma \sigma (T_s^4 - T_\infty^4)$$

$$q = q'' A \quad \sigma \triangleq 5.67 \times 10^{-8} \left[\frac{W}{m^2 K^4} \right] \quad E = \Sigma \sigma T_s^4$$

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st} \quad \alpha \equiv \text{therm diffusivity} \left[\frac{m^2}{s} \right]$$

$$\Sigma q_{in} - \Sigma q_{out} + \dot{q} = \rho V c_p \frac{\partial T}{\partial t} \quad \alpha = \frac{k}{\rho c_p}$$

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

Cylindrical

$$q'' = -k \vec{\nabla} T = -k \left(\hat{i} \frac{\partial T}{\partial x} + \hat{j} \frac{1}{r} \frac{\partial T}{\partial y} + \hat{k} \frac{\partial T}{\partial z} \right)$$

$$\rho c_p \frac{\partial T}{\partial t} = \left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q} \right\}$$

Spherical

$$q'' = -k \vec{\nabla} T = -k \left(\hat{i} \frac{\partial T}{\partial r} + \hat{j} \frac{1}{r} \frac{\partial T}{\partial \theta} + \hat{k} \frac{1}{r \sin(\theta)} \frac{\partial T}{\partial \phi} \right)$$

$$\rho c_p \frac{\partial T}{\partial t} = \left\{ \frac{1}{r^2} \frac{\partial}{\partial r} \left(k r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(k \sin(\theta) \frac{\partial T}{\partial \theta} \right) + \dot{q} \right\}$$

Thermal Resistance $\dot{q} = 0$

	Conduction	Convection	Radiation
Plane Wall	$R_t = \frac{L}{kA}$	$R_t = \frac{1}{hA}$	$R_t = \frac{1}{h_R A}^*$
Cylindrical	$\ln \left(\frac{r_2/r_1}{2\pi L k} \right)$		
Spherical	$\left(\frac{1}{r_1} - \frac{1}{r_2} \right) / 4\pi k$		

$$^* h_R = \varepsilon \sigma (T_s + T_{sur})(T_s^2 + T_{sur}^2)$$

$$R_{t,contact} = \frac{T_A - T_B}{q_x} \quad R''_{t,contact} = \frac{T_A - T_B}{q''_x}$$

$$R_{tot} = \Sigma R_t \quad q_x = \frac{T_{bigger} - T_{smaller}}{R_{tot,between}}$$

Alternate Conduction $\dot{q} = 0$

- Plane Wall

$$q_x = \int_{x_0}^{x_f} \frac{dx}{A(x)} = - \int_{T_0}^{T_f} k(T) dT$$

Heat rate and heat flux are constant.

- Cylindrical

$$q_r = -k(2\pi r L) \frac{dT}{dr} = \frac{T_{s1} - T_{s2}}{\frac{\ln(r_2/r_1)}{2\pi L k}}$$

Heat rate is constant. Heat flux is **NOT**

$$T(r) = \frac{T_{s1} - T_{s2}}{\ln(r_2/r_1)} \ln \left(\frac{r}{r_2} \right) + T_{s2}$$

- Spherical

$$q_r = \frac{T_{s1} - T_{s2}}{\left(\frac{1}{r_1} - \frac{1}{r_2} \right) / 4\pi k} = -4k\pi r^2 \frac{dT}{dr}$$

Heat rate is constant. Heat flux is **NOT**

Fins

1D, steady-state, $\dot{q} = 0$, constant properties

$$\frac{d^2 T}{dx^2} + \left(\frac{1}{A_c} \frac{dA_c}{dx} \right) \frac{dT}{dx} - \left(\frac{1}{A_c} \frac{h dA_s}{k dx} \right) (T - T_\infty) = 0$$

if const. $A_c \bigwedge \theta(x) \triangleq T(x) - T_\infty$ then:

$$\frac{d^2 \theta}{dx^2} - m^2 \theta(x) = 0 \quad m = \sqrt{\frac{hP}{kA_c}}$$

$$\varepsilon_f = \frac{q_f}{hA_{c,b}\theta_b} = \frac{\text{fin heat rate}}{\text{heat rate w/o fin}} = \sqrt{\frac{kP}{hA_c}}$$

$$\varepsilon_f = \frac{R_{t,b}}{R_{t,f}} = \frac{\text{resist. @ base w/o fin} \because \text{conv.}}{\text{max heat rate if all rod @ } T_b}$$

$$\eta_f = \frac{q_f}{q_{max}} = \frac{q_f}{hA_{f,\theta_b}} = \frac{\text{fin heat rate}}{\text{max rate if all rod @ } T_b}$$

$$\eta_f = \frac{\tanh(L_c)}{mL_c} \quad \square \rightarrow L_c = L + t/2 \quad \bigcirc \rightarrow L_c = L + D/4$$

Finite Difference Equations

1. Pick the node you care about, it is node (m,n)
2. Determine q from surrounding nodes (pointed towards node m,n)
3. Do an energy balance, solve for $T_{(m,n)}$

Lumped Capacitance Method

Check Biot number first, if $Bi < 0.1$, L.C.M. is valid.

$$Bi \triangleq \frac{hL_c}{k} \quad L_{c,plane} = L \quad L_{c,cyl} = \frac{r_o}{2} \quad L_{c,sph} = \frac{r_o}{3}$$

$$t = -\frac{\rho V c}{h A_s} \ln \left(\frac{\theta}{\theta_i} \right) \quad \tau_t = \rho V c \frac{1}{h A_s} = C_t R_t \quad L_c = \frac{V}{A_s}$$

$$\frac{\theta}{\theta_i} = \frac{T - T_\infty}{T_i - T_\infty} = \exp \left(-\frac{h A_s}{\rho V c} t \right) = \exp(-Bi \cdot Fo)$$

$$Fo = \frac{\alpha t}{L_c^2} = \frac{kt}{\rho c_p L_c^2} \quad Q = \rho V c (T_i - T_\infty) (1 - e^{-t/\tau}) [J]$$

Spatial Effects

- If $Bi \not< 0.1$, take into account spatial effects (Use the Bi formula listed at the bottom of the table).
- Assume $Fo > 0.2$, to use the single term approximation. If $Fo \not> 0.2$, the infinite series must be dealt with. **Good luck.**

$$\theta^* \triangleq \frac{\theta}{\theta_i} = \frac{T - T_\infty}{T_i - T_\infty} \quad 0 \leq \theta^* \leq 1 \quad \frac{\partial^2 \theta^*}{\partial x^{*2}} = \frac{\partial \theta^*}{\partial Fo}$$

$$x^* \triangleq \frac{x}{L} \quad 0 \leq x^* \leq 1 \quad t^* = Fo = \frac{\alpha t}{L_c^2}$$

$$\theta^* = \sum_{n=1}^{\infty} C_n \exp(-\zeta_n^2 Fo) \cos(\zeta_n x^*) \quad C_n = \frac{4 \sin(\zeta_n)}{2\zeta_n + \sin(2\zeta_n)}$$

Approximate solution (1-term) is only valid for symmetric, or adiabatic on one side

$$\theta_0^* = C_1 \exp(-\zeta_1^2 Fo) = \frac{T - T_\infty}{T_i - T_\infty} \quad (\theta_0^* \text{ is midplane of } \theta^*)$$

$$\theta^* = \theta_0^* \cos(\zeta_1 x^*)$$

Locations of Appendices and Tables

Fin temp table	Table 3.4	pg 161
Finite Diff. Equations	Table 4.2	pg 246
Spatial effects coeff.	Table 5.1	pg 301
Thermophysical props.	Appendix A	pg 981
Zeta values for $\zeta_1 - \zeta_4$	Appendix B.3	pg 1016
Bessel Functions	Appendix B.4	pg 1017
Heat generation eq's	Appendix C	pg 1019

TABLE 3.4 Temperature distribution and heat loss for fins of uniform cross section

Case	Tip Condition ($x = L$)	Temperature Distribution θ/θ_b	Fin Heat Transfer Rate q_f
A	Convection heat transfer: $h\theta(L) = -k d\theta/dx _{x=L}$	$\frac{\cosh m(L-x) + (h/mk) \sinh m(L-x)}{\cosh mL + (h/mk) \sinh mL}$ (3.75)	$M \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL}$ (3.77)
B	Adiabatic: $d\theta/dx _{x=L} = 0$	$\frac{\cosh m(L-x)}{\cosh mL}$ (3.80)	$M \tanh mL$ (3.81)
C	Prescribed temperature: $\theta(L) = \theta_L$	$\frac{(\theta_L/\theta_b) \sinh mx + \sinh m(L-x)}{\sinh mL}$ (3.82)	$M \frac{(\cosh mL - \theta_L/\theta_b)}{\sinh mL}$ (3.83)
D	Infinite fin ($L \rightarrow \infty$): $\theta(L) = 0$	e^{-mx} (3.84)	M (3.85)

$$\theta = T - T_\infty \quad m^2 = hPk/A_s \quad \theta_b = \theta(0) = T_b - T_\infty \quad M = \sqrt{hPkA_s} \theta_b$$

TABLE 4.2 Summary of nodal finite-difference equations

Configuration

Finite-Difference Equation for $\Delta x = \Delta y$

Case 1. Interior node

$$T_{m,n+1} + T_{m,n-1} + T_{m-1,n} + T_{m+1,n} - 4T_{m,n} = 0$$

$$2(T_{m-1,n} + T_{m,n+1}) + (T_{m+1,n} + T_{m,n-1}) + 2\left(\frac{h\Delta x}{k} T_\infty - 2\right) \left(3 + \frac{h\Delta x}{k}\right) T_{m,n} = 0$$

Case 2. Node at an internal corner with convection

$$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2h\Delta x}{k} T_\infty - 2\left(\frac{h\Delta x}{k} + 2\right) T_{m,n} = 0$$

Case 3. Node at a plane surface with convection

$$(T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + 2\frac{h\Delta x}{k} T_\infty - 2\left(\frac{h\Delta x}{k} + 1\right) T_{m,n} = 0$$

Case 4. Node at an external corner with convection

$$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2h\Delta x}{k} T_\infty - 4T_{m,n} = 0$$

Case 5. Node at a plane surface with uniform heat flux

$$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2q''\Delta x}{k} - 4T_{m,n} = 0$$

TABLE 5.1 Coefficients used in the one-term approximation to the series solutions for transient one-dimensional conduction

Bi^*	Plane Wall		Infinite Cylinder		Sphere	
	ζ_1 (rad)	C_1	ζ_1 (rad)	C_1	ζ_1 (rad)	C_1
0.01	0.0998	1.0017	0.1412	1.0025	0.1730	1.0030
0.02	0.1410	1.0033	0.1995	1.0050	0.2445	1.0060
0.03	0.1723	1.0049	0.2440	1.0075	0.2991	1.0090
0.04	0.1987	1.0066	0.2814	1.0099	0.3450	1.0120
0.05	0.2218	1.0082	0.3143	1.0124	0.3854	1.0149
0.06	0.2425	1.0098	0.3438	1.0148	0.4217	1.0179
0.07	0.2615	1.0114	0.3709	1.0173	0.4551	1.0209
0.08	0.2791	1.0130	0.3960	1.0197	0.4860	1.0239
0.09	0.2956	1.0145	0.4195	1.0222	0.5150	1.0268
0.10	0.3111	1.0161	0.4417	1.0246	0.5423	1.0298
0.15	0.3779	1.0237	0.5376	1.0365	0.6609	1.0445
0.20	0.4328	1.0311	0.6170	1.0483	0.7593	1.0592
0.25	0.4801	1.0382	0.6856	1.0598	0.8447	1.0737
0.30	0.5218	1.0450	0.7465	1.0712	0.9208	1.0880
0.4	0.5932	1.0580	0.8516	1.0932	1.0528	1.1164
0.5	0.6533	1.0701	0.9408	1.1143	1.1656	1.1441
0.6	0.7051	1.0814	1.0184	1.1345	1.2644	1.1713
0.7	0.7506	1.0919	1.0873	1.1539	1.3525	1.1978
0.8	0.7910	1.1016	1.1490	1.1724	1.4320	1.2236
0.9	0.8274	1.1107	1.2048	1.1902	1.5044	1.2488
1.0	0.8603	1.1191	1.2558	1.2071	1.5708	1.2732
2.0	1.0769	1.1785	1.5994	1.3384	2.0288	1.4793
3.0	1.1925	1.2102	1.7887	1.4191	2.2889	1.6227
4.0	1.2646	1.2287	1.9081	1.4698	2.4556	1.7202
5.0	1.3138	1.2402	1.9898	1.5029	2.5704	1.7870
6.0	1.3496	1.2479	2.0490	1.5253	2.6537	1.8338
7.0	1.3766	1.2532	2.0937	1.5411	2.7165	1.8673
8.0	1.3978	1.2570	2.1286	1.5526	2.7654	1.8920
9.0	1.4149	1.2598	2.1566	1.5611	2.8044	1.9106
10.0	1.4289	1.2620	2.1795	1.5677	2.8363	1.9249
20.0	1.4961	1.2699	2.2881	1.5919	2.9857	1.9781
30.0	1.5202	1.2717	2.3261	1.5973	3.0372	1.9898
40.0	1.5325	1.2723	2.3455	1.5993	3.0632	1.9942
50.0	1.5400	1.2727	2.3572	1.6002	3.0788	1.9962
100.0	1.5552	1.2731	2.3809	1.6015	3.1102	1.9990
∞	1.5708	1.2733	2.4050	1.6018	3.1415	2.0000

* $Bi = hL/k$ for the plane wall and hr_o/k for the infinite cylinder and sphere. See Figure 5.6.